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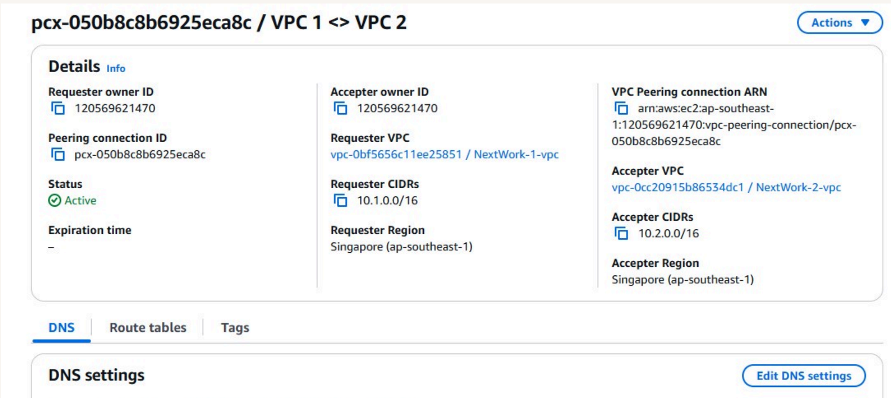
VPC Peering

Sayed Mahmeer Ali Shah



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a private cloud which provides an isolated network.

How I used Amazon VPC in this project

I used two VPCs and also did peer connection between them.

One thing I didn't expect in this project was...

I did not expect the errors but finally after some changes I successfuly connected them both.

This project took me...

It took me almost 50 minutes to complete this project due to some errors.

In the first part of my project...

Step 1 - Set up my VPC

In this step, we are going to create VPCs.

Step 2 - Create a Peering Connection

In this step we are going to establish a connection between the VPC (peering).

Step 3 - Update Route Tables

In this step, we are going to set up a way for traffic coming from VPC 1 to get to VPC 2, and set up a way for traffic coming from VPC 2 to get to VPC 1.

Step 4 - Launch EC2 Instances

In this step we are going to launch an EC2 Instance to test peering connection.

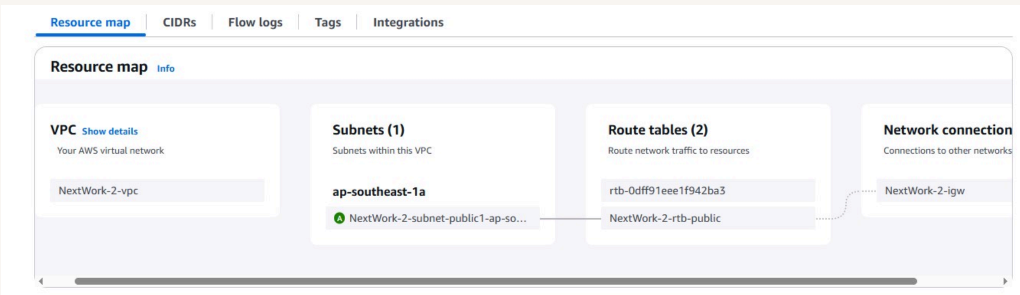
Multi-VPC Architecture

I started my project by launching a VPC and more with 1 public subnet and 0 private subnets.

The CIDR blocks for VPCs 1 and 2 are different. They have to be unique because if they are the same, they can create routing conflicts or connectivity issues.

I also launched 2 EC2 instances

I didn't set up key pairs for these EC2 instances, as proceeding without a key pair means AWS will manage it for us, so we don't have to worry about.

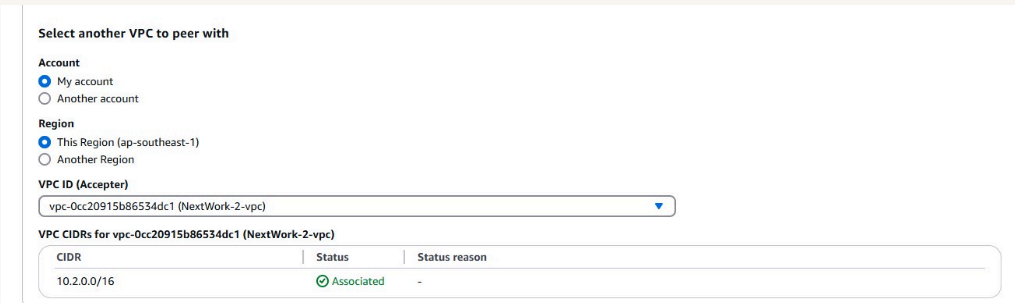


VPC Peering

A VPC peering connection is a service that connects two different VPCs.

VPCs would use peering connections to communicate and connect with one another on different networks.

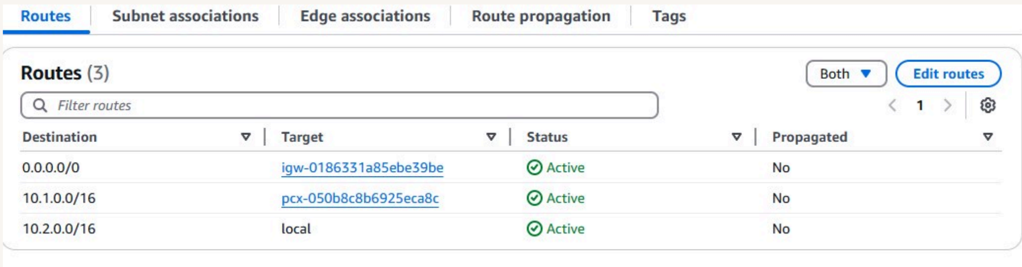
The difference between a Requester and an Accepter in a peering connection is: a requester request the connection for a VPC and accepter accepts the connection request of the other VPC.



Updating route tables

After accepting a peering connection, my VPCs' route tables need to be updated because I have to allow them to peer with one another, establishing a connection via route table is the key.

My VPCs' new routes have a destination of peering connection. The routes' target was to establish connection with the other VPC.



In the second part of my project...

Step 5 - Use EC2 Instance Connect

In this step we use EC2 Instance Connect to connect to your first EC2 instance, and fix a connection error!

Step 6 - Connect to EC2 Instance 1

in this step we use EC2 Instance Connect to connect to Instance 1 (one more time), and fix (another) error.

Step 7 - Test VPC Peering

in this step we get Instance 1 to send test messages to Instance 2. Solve connection errors until Instance 2 is able to send messages back.

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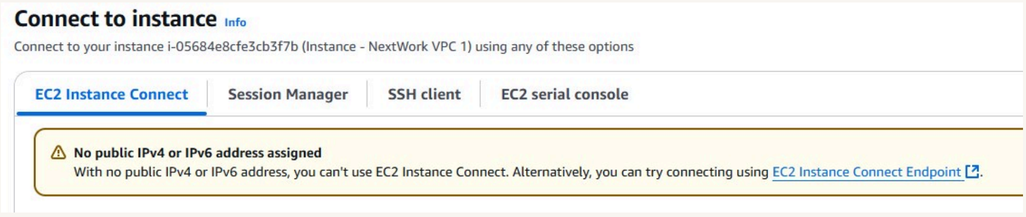
Troubleshooting Instance Connect

Next, I used EC2 Instance Connect to establish a connection between both Ec2s.

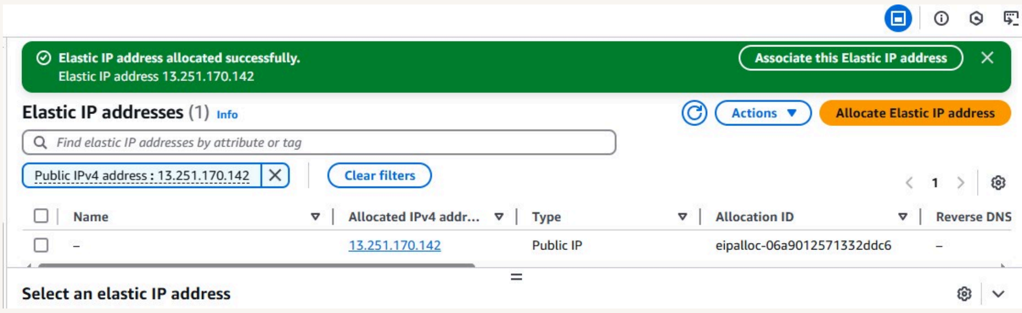
I was stopped from using EC2 Instance Connect as Ec2 did not have any public ip before.



Elastic IP addresses

To resolve this error, I set up Elastic IP addresses. Elastic IP addresses are static, public IPv4 addresses designed specifically for the dynamic nature of cloud computing.

Associating an Elastic IP address resolved the error because it provided us an IP to access for.

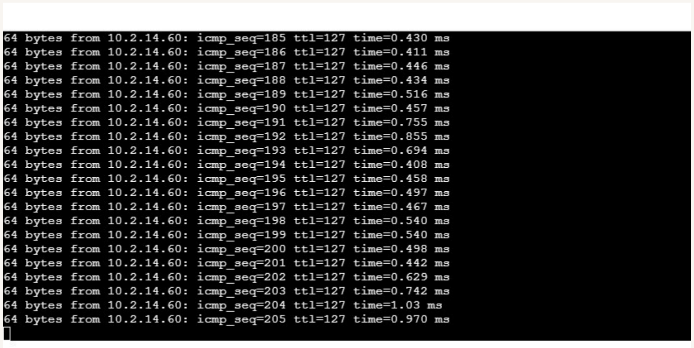


Troubleshooting ping issues

To test VPC peering, I ran the command ping.

A successful ping test would validate my VPC peering connection because i made all the necessary changes to security group inbound rules.

I had to update my second EC2 instance's security group because to establish connection. I added a new rule that allowed SSH and ICMP



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